

How Google Crawls in the Age of AI SEARCH

A senior-level guide to Day 1 in Barcelona: what Google showed about AI Mode, how the crawler really works, why robots.txt breaks, and how crawl budget is decided, plus the community talks and the Q&A. Every point labelled and checked against Google's documentation.

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NOTES & GUIDE BY

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Deep Dive edition · an independent attendee guide

DIVE LOG • DAY 1 FIELD GUIDE

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HOW TO READ THE LABELS

SLIDE Shown on screen during the session

STAGE Said on stage, from my notes or a recording

DOCS Google's published documentation

PRESS A third-party report of what Google said

ANALYSIS My interpretation, not Google's words

(*not in docs*) Said at the event; not in Google's docs

SUMMARY

Key takeaways

AI features run on the same crawl and index as classic Search. Whether Google reaches and fetches your pages still decides everything downstream, including AI answers.

01

Good SEO is good "GEO".

AI Overviews and AI Mode are built on the core ranking systems and grounded in the Search index. No separate discipline to buy.

02

Expect traffic to move.

Google said outright that long-held traffic patterns are likely to fluctuate. Growth is in brainstorming, "which" and planning queries.

03

There is now a real AI opt-out.

A Search Console setting removes a property from AI Overviews, AI Mode and AI in Discover without, Google says, affecting regular rankings.

04

llms.txt does nothing for Google Search.

Google's guide says such files "neither harm nor help"; Gary Illyes doesn't expect that to change.

05

One crawling platform serves Google's products.

Googlebot is one client of it. Search, Ads, Shopping and Images share it, and one hostload limit per host.

06

Your server's health sets the crawl rate.

Connect time, time to first byte and 429 or 5xx responses decide how hard Google crawls. Firewalls that silently block Googlebot look like a failing server.

07

Login walls and paywalls read as 404.

Pages that answer Google's crawlers with a 403 or a 402, both on the rise, are dropped. A CDN captcha served with 200 becomes a soft 404.

08

A crawler obeys exactly one robots.txt group.

Groups don't add up: Googlebot ignores the `*` group when a Googlebot group exists, the longest matching rule wins, and one stray line can merge or split groups.

09

Crawl demand is inherited by path.

With little data on a URL, Google uses the quality and popularity of its parent folder, then that folder's parent.

10

noindex costs crawl; disallow doesn't.

Google must fetch a page to read noindex. crawl-delay is ignored. Faceted URLs are the classic budget leak.

11

AI Mode doesn't mean more Googlebot.

AI Mode sits on the existing crawl infrastructure, and crawling more often doesn't make a page rank better.

12

There is no duplicate content penalty.

Google picks one URL to keep. The real cost is losing control of which one, and crawling every copy.

01

SECTION 01

Day 1 at a glance

The day followed one arc: the state of Search, then the crawl pipeline, then the three ways sites waste or block crawling — errors, robots.txt and crawl budget. Community lightning talks and a Q&A ran in between.

11:00	 Welcome and opening keynotes Lino Cattaruzzi, President, Google Iberia	Slides · Transcript
11:35	 What's new in the world of Search Gary Illyes, Search Relations	Transcript · Slides
11:45	 How Search works and where's AI? Cherry Prommawin and Gary Illyes, Search Relations	Transcript · Slides
13:10	 Lightning session A: Automation and AI James Powley, Rafael Kovashikawa, Kira Breuer, Carlos Ortega, Thiago Pojda and other community speakers	Transcript
14:05	 How crawling works Cherry Prommawin and Gary Illyes, Search Relations	Slides · Transcript
14:35	 How crawling errors affect Search Cherry Prommawin and Gary Illyes, Search Relations	Notes only · Transcript
15:10	 How Google interprets robots.txt Google	Slides · Transcript
15:30	 Lightning session B: Robots.txt Dave Smart and other community speakers	Transcript
15:40	 What would you do?: Crawling Edition Google	Not covered
16:00	 How Google thinks about crawl budget Cherry Prommawin, Search Relations	Slides · Transcript
16:20	 Lightning session C: Crawling Tobias Schwarz and Jovana Avramovic	Transcript
16:35	 Q&A John Mueller and Gary Illyes, Search Relations	Transcript
17:25	 Daily wrap-up Google	Transcript

This edition adds a second attendee's recording of most of the day and six audio recordings to my slide photos and notes; where they disagree, the audio wins. Still not covered: "What would you do?" and the rest of Lightning session B.

DAY 1 · TODAY'S GUIDE

Crawling

Discovery, fetching, robots.txt, crawl budget

DAY 2

Indexing

One big index, signals, more than words, BERT

DAY 3

Serving

Ranking, AI features, grounding, query fan-out

The three-day arc. The event followed a page through Google Search, one stage per day, as shown in "How Search works".

02

SECTION 02 • KEYNOTE AND "WHAT'S NEW"

The state of Search

Google's message: AI is growing total search use rather than replacing it, and traffic will shift as a result.



"Search is never a solved problem. Old challenges evolve and new challenges are constantly popping up — because the internet and the world are always changing."

ELIZABETH REID, VP, SEARCH — QUOTED ON A KEYNOTE SLIDE

SLIDE The theme guiding the future of Search: **"Making it effortless to find helpful information."** Three forces were named as drivers: new platforms (closed-garden apps taking information needs away from the open web), new user preferences (authentic, visual, easy-to-scan content) and new technologies (AI).

STAGE Lino Cattaruzzi, opening keynote: "Google Search now is AI search." The new intelligent search box is the biggest change to it in 25 years; it takes text, speech or images, and AI Overviews and AI Mode now work as one experience. He added that YouTube has passed Netflix as the number one streaming service, in the US and around the world. **ANALYSIS** Cite Nielsen's US watch time: nothing published backs "around the world".

○ The ground rules: Google doesn't sell rankings

STAGE The host opened with Google's **Honest Results Policy**: no payment buys an organic ranking, ads are marked as sponsored, and no client, partner or personal contact gets an advantage. So everything shared at the event is information available to everyone.

○ The numbers Google showed

Google-reported figures: two rows from the slides (footnotes cite Google internal data and Google-commissioned Ipsos surveys; only the follow-up and brainstorming figures were marked US), and a last row said on stage and checked against Google's posts. Google's framing, not independent research.

5T+

searches on Google
per year

2× / qtr

AI Mode query volume,
doubling every quarter
since launch

~3×

length of an AI
Mode query vs. a
traditional one

1 in 6

AI Mode searches are
multimodal (voice
or image)

+40%

follow-up queries in
AI Mode, per month
on average
US, since launch

1.5×

faster volume growth for
queries of 5+ words
Global English queries

94%

of frequent LLM users
are also frequent
Google users

71%

of shoppers on Google
are open to new brands
Ipsos survey

1B+

monthly active users of AI Mode alone

2.5Bpeople using Google's AI search
I/O 2026: AI Overviews, monthly**1B+ / wk**

clicks to the web from AI Overviews and AI Mode AI responses alone

1.5Bpeople use Google Lens to search what they see
Monthly, May 2025 post

STAGE Google also said it sends billions of clicks to the open web every day, and that 77% of AI Overviews and AI Mode users decide faster, with 18 to 24 year olds asking more, and more complex, questions (*the last two not in docs*).

○ Four new ways people search in AI Mode

EXPLORE

Brainstorming queries growing **30% faster** than AI Mode overall in the US since launch. "Where to", "ideas for", "where should I".

LEARN

Study guides and quizzes, topic deep dives, professional development.

DECIDE

Queries starting with "which" growing **40% faster** than AI Mode queries overall over the last six months.

DO

Planning queries — restaurants, itineraries, finances, fitness — growing **80% faster** than AI Mode queries overall over the last six months.

SLIDE Google also showed **Search Agents** in AI Mode: standing requests such as "tell me if Samsung Frame 2000 goes on sale". **STAGE** AI Mode turns a plain-language request into an alert; Gary Illyes said they had launched globally a day or two before (*not in docs*). **PRESS** A search agency reported the rollout, announced on 28 September.

○ Google's five principles for the ecosystem

#	PRINCIPLE	WHAT GOOGLE SAID SLIDE
1	Consistent testing	Features are tested, then woven into core Search when ready.
2	The SERP will evolve	Beyond organic links and ads: videos, cards and more.
3	Multiple user needs	Some want links, some quick answers, some depth. Google says it will keep prioritising traffic to the ecosystem.
4	Incentivise quality	"Made for Search" content will not be successful.
5	Traffic patterns may fluctuate	Long-held patterns are likely to change, "creating new opportunities for all sites".

STAGE On principle 4, Lino Cattaruzzi: "incentivise high-quality content, created by humans or by AI. But the key is that it's high quality". His close: "UEO", user engine optimisation.

ANALYSIS

Principle 5 is the line to plan around: Google told a room of SEOs to expect traffic to move. Growth is in brainstorming, "which" and planning queries, which a one-keyword page fits poorly. Build pages that help someone decide.

03

SECTION 03 • WHAT'S NEW

New publisher controls

Google listed eight ways AI Search sends people to websites, and separately showed a new opt-out control for generative AI. The opt-out is the one most site owners can act on today.

SLIDE The full list on screen: preferred sources expansion, fresh perspectives and prominent links, subscription content, in-line quotes in AI Mode and AI Overviews, rich linking and embedded web, context on content sources, "Highly Cited" labels, and Search profiles.

The generative AI opt-out in Search Console

SLIDE "Opt-out control for generative AI: direct control over AI Overviews and AI Mode without impacting web rankings." The setting is called **Search generative AI** and has three states.

Search generative AI

Property setting • Search Console

- ☐ **Inherit from parent property**
Default. Follows the closest parent that changed it
- ☒ **Include**
Links and content can appear and ground AI answers
- ☐ **Exclude**
No links, no grounding, no AI impressions or traffic

IT CONTROLS

- ✓ AI Overviews
- ✓ AI Mode
- ✓ Generative AI features in Discover

IT DOES NOT TOUCH

- ✗ Regular rankings or inclusion
- ✗ Gemini training (use Google-Extended)
- ✗ Individual pages (property level only)

What the switch does. Illustration based on the slide and Google's Search Console help page; not a screenshot of the product.

DOCS Google's help page adds the operational details. The control "isn't used as a ranking or inclusion signal" for regular results. Changes generally apply within 1–2 days, though some content takes longer. A child property inherits from its closest parent that changed the setting, unless you override it.

How to decide **ANALYSIS**

YOUR SITUATION	LIKELY CHOICE	WHY
Lead generation, local services, e-commerce	Include	AI answers cite and link sources; excluding removes you from a growing surface.
Paywalled or licensed content you sell	Test exclude on a child property	Protects the asset, but you lose AI citations. Measure before rolling out.
You only want to stop Gemini training	Include + Google-Extended	The opt-out does not control training.
You want out of AI and regular snippets	nosnippet	Wider effect: it also removes normal snippets.

Before changing anything, read the **Generative AI performance report** in Search Console. **DOCS** Google's launch post (3 June 2026) lists impressions, pages, countries, devices (Search only) and dates, and no click or query metrics — it measures exposure, not value. **STAGE** Gary Illyes said the reporting starts with impressions only and may expand later; the Q&A covered why query data is hard.

○ Search profiles, preferred sources and subscriptions

SEARCH PROFILES

SLIDE A claimable profile: photo, bio, social accounts, pinned posts, links. **DOCS** Launched 4 June 2026, US first. Since 16 September 2026, US publishers and creators qualify with 10,000 followers across YouTube, Instagram, X or TikTok.

PREFERRED SOURCES

SLIDE Sites users mark as preferred get more visibility in Top Stories and a label in AI Mode and AI Overviews.

LINKED SUBSCRIPTIONS

SLIDE Users link subscriptions so answers highlight paywalled content they already pay for. One publisher saw 34% more engagement.

ANALYSIS

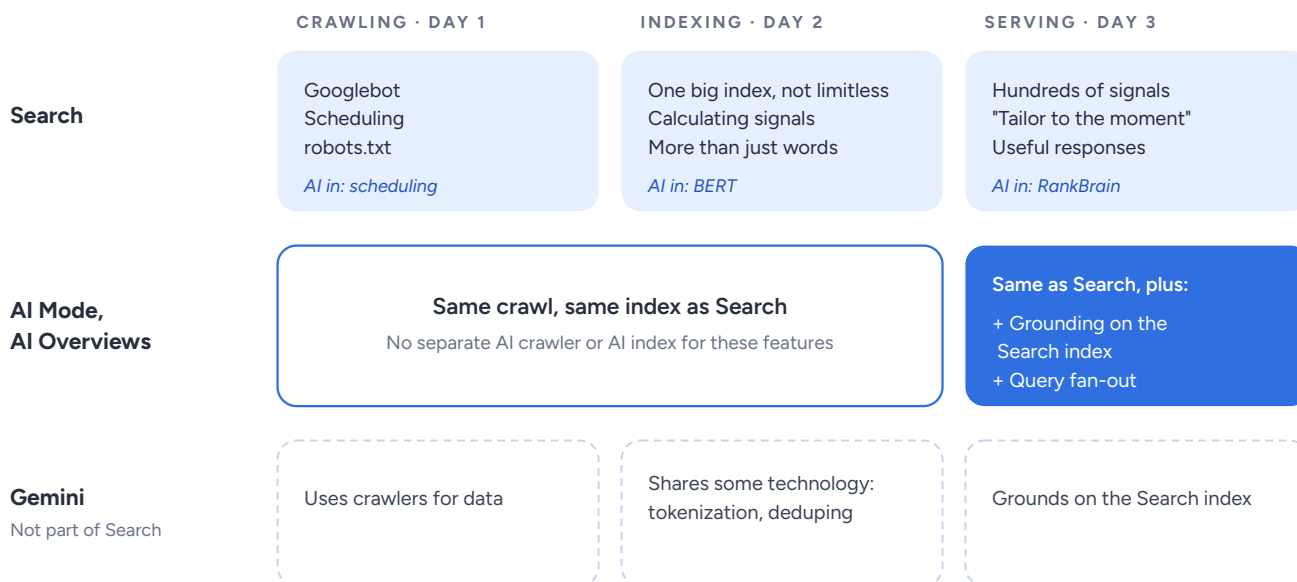
All three depend on users acting on your behalf. That rewards brands with a loyal audience and does little for anonymous lead-generation sites. Brand building is now ranking-adjacent work, not only marketing.

04

SECTION 04 · HOW SEARCH WORKS AND WHERE'S AI?

How Search works, and where AI sits

"If you know how Search works, you largely also know how AI features work." AI Mode and AI Overviews use the same crawl and index as classic Search. They differ only at serving.



One pipeline, three products. Redrawn from the session's matrix slide, which also served as the roadmap for the three days.

STAGE Gary Illyes: AI Overviews and AI Mode may have extra processes of their own, like any search feature, but the bulk of their processing is the same as for normal Search.

The pipeline, step by step

STEP	WHAT CHERRY PROMMAWIN EXPLAINED STAGE
Discovery	There are trillions of URLs, so many that even Google can't count them, and some will never be found (<i>not in docs</i>). Hub pages such as the homepage and categories may be visited more often, because they link to new pages.
Crawling	A crawler downloads a page, extracts its links and repeats; Googlebot is Search's main crawler.
Indexing	The HTML is parsed, JavaScript-heavy pages are rendered with the latest Chromium, duplicates are clustered under one canonical URL, and signals are stored for index selection and later for ranking.
Index selection	Decides what goes into an index that is big but not limitless; a selected page takes its whole duplicate cluster with it. Printed, the index would reach the Moon and back twelve times (<i>not in docs</i>).
Serving	Happens in real time: the query is cleaned up, its language detected and expanded; quality is one of the most important ranking factors; then the results page is built from the features that fit the intent. A majority of search features now use AI (<i>not in docs</i>).

○ The AI already inside Search is not new

INDEXING

Statistical models

SLIDE Used for 20+ years: spam detection, and originally "Did you mean".

INDEXING

BERT

SLIDE Reads each word in the context of the whole sentence.

DOCS Google's ranking systems guide lists it as a ranking system.

SERVING

RankBrain

SLIDE Interprets the intent behind new or unusual queries.

SERVING

MUM

SLIDE Works across text, images, audio and video in 75+ languages.

DOCS The 2021 launch post named text and images, with video and audio to come; the ranking systems guide says MUM is not used for general ranking.

STAGE Gary Illyes dated "Did you mean" to around 2001–2002, a statistical model he counts as AI, and RankBrain to around 2015–2016, the first AI system Google talked about widely. MUM reads context: hiking shoes for Mount Everest, not Kilimanjaro. **ANALYSIS** MUM was announced in May 2021, more than five years after RankBrain; date these systems by Google's posts.

SLIDE Ranking signals differ by result type: **web pages** (text, links, passages), **images** (resolution, colour, associated text), **news** (freshness, originality, diversity), **local** (location, type, rating, reviews, hours) and **video** (language, text from speech).

○ "Good SEO is good GEO"

STAGE Gary Illyes argued that "GEO" is a new label invented to create a new field. The work doesn't need it: understand how SEO works and you are covered. He traced "SEO is dead" posts back to a forum thread called "Search engine R.I.P." from 10 November 1997; they still appear every month. SEO is not dead, he said: the AI features create more opportunities.

SLIDE "Don't worry about what to call it. Good SEO is good GEO, AEO." Three reasons: generative AI features are built on the core ranking systems; query fan-out expands the original query to find related information; and generative AI features highlight content indexed by Google Search.

DOCS Google's optimisation guide for generative AI (updated 10 July 2026) agrees: optimising for generative AI search "is optimizing for the search experience, and thus still SEO". It describes query fan-out as running several related searches at once — a question about lawn weeds also searches herbicides and prevention. **STAGE** Gary Illyes called fan-out nothing new: "just ten different searches that fire in the background", each a normal search on the same systems.

○ Four myths Google addressed

MYTH	GOOGLE'S ANSWER SLIDE
Optimise for AI over readers	Optimise for people. No need to obsess over keywords or AI phrasing, no need to chop content, "LLMS.txt isn't necessary".
Build content for every possible need	Prioritise unique perspectives and expertise beyond common knowledge.
Your website is no longer relevant	Keep content accessible: crawlable, well structured, fast and easy to read, for readers and AI tools.
Old metrics don't work in the AI era	Focus on real business goals and measure what matters to your business.

GOOGLE'S "YOUR QUESTION" SLIDE • "CAN GOOGLE DISTINGUISH BETWEEN AI AND HUMAN WRITTEN CONTENT?"

SLIDE "ML based ranking algorithms and signals are trained on content by humans for humans. They 'understand' and promote natural content better." Gary Illyes presented it as the question this topic usually prompts, not one asked in the room.

STAGE Google is not really trying to tell the two apart: "We care more about the quality of the content than how it was created." The natural content its systems favour includes content created by humans "or at least edited and reviewed" by them (*that part not in docs*).

ANALYSIS This is not a claim that Google detects AI text. It says ranking models favour text that reads as natural to people. Google's policy is consistent: AI-assisted content is allowed, while mass-produced pages made mainly to rank fall under its scaled content abuse policy. The risk is in scale without value, not in the tool.

○ Searching with images

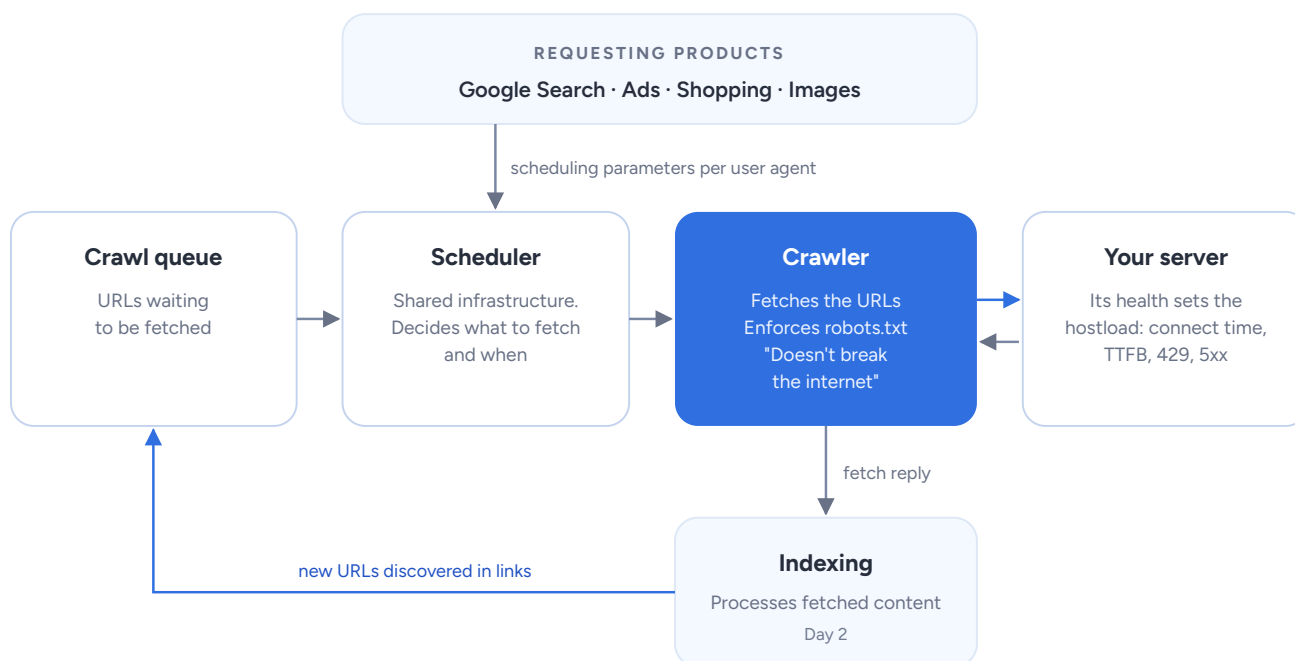
- **STAGE** Google noticed young users uploading pictures to reverse image search and expecting answers about the picture; Google Lens was its answer, now used by over 1.5 billion people (*the first part not in docs*).
- **STAGE** For visual search, Google breaks an image down into vectors, sends them to the index and returns results based on them (*not in docs*).
- **STAGE** Gary Illyes: searching by image or video, as Search offers it today, wasn't possible before transformers (*not in docs*). **ANALYSIS** Google dates Google Images to 2001 and Lens to 2017, so read this as being about today's multimodal AI search. For image work, see "Images matter" in the From the floor section.

05

SECTION 05 • CHERRY PROMMAWIN AND GARY ILLYES

How crawling works

Google runs one crawling system for all its products, and its first job is not to harm your server. That is why speed and errors on your side decide how much Google crawls.



Google's crawl pipeline. Redrawn from the "How crawling works" and crawl budget slides.

- **SLIDE** The crawler has "multiple tasks": fetch from the internet, "ensure we don't break the internet", and enforce robots.txt policies. Fetched data goes on to indexing.
- **SLIDE** The scheduler is "shared scheduler infrastructure". It decides what to fetch and when; each team sets the scheduling parameters for its own user agent.
- **SLIDE** Search, Ads, Shopping and Images all request through one "centralized crawling infrastructure". Its primary mandate: fetch "while strictly preventing the overloading of external servers".
- **SLIDE** Discovery runs through links: a news homepage links to `/sport` and `/tech`, which link to their articles.

Googlebot is just an HTTP client

STAGE Gary Illyes: "Googlebot is just a client. It is an HTTP client. There's nothing all that much special about it." Like a browser, it fetches the URL it was given and returns the bytes. Google's crawling is "a large-scale distributed swarm of simple HTTP clients", and AI agents are technically the same thing: HTTP clients acting for a user or a service (*not in docs*).

STAGE Google runs probably hundreds, if not thousands, of crawlers (*not in docs*), some named and some not, all on one infrastructure that enforces the same crawling policies. Googlebot is the crawler for web search, AI features included. **DOCS** Google's Inside Googlebot post speaks of dozens of other clients, with only the larger ones documented. **ANALYSIS** A Google user agent missing from the public lists is not proof of a fake: verify by reverse DNS or Google's published IP ranges.

DOCS Googlebot fetches only the first 2MB of each URL, HTTP headers included (64MB for PDFs). Anything past the cutoff is not fetched, rendered or indexed, so keep the title, canonical and key structured data high in the HTML.

○ How the scheduler decides

- **STAGE** Google finds URLs mainly in pages it has crawled, and also in sitemaps: a format from around 2005, "nothing fancy", that Google still uses.
- **STAGE** The scheduler logs how often each page changes and crawls fast-changing pages first: a news homepage before a terms of service page that changes once a year.
- **STAGE** URLs or sites known to be historically spammy are very likely deprioritised.
- **STAGE** Each team sets its own priorities. Search cares a lot about quality; Ads checks every publisher page that wants to show ads, as the URLs come in. Crawling for Gemini may care more about amount, because for language models "the number of tokens is actually more important" than the quality (*not in docs*).

STAGE To debug crawling, Gary Illyes called Search Console's **Crawl Stats** report (Settings → Crawl stats) the most valuable tool: how much Google crawls, the errors it got and the content types each crawler fetched. Server logs are "the real thing", for those who can read them.

ANALYSIS · TWO CONSEQUENCES

Orphan pages are slow pages. A URL with no internal links depends on sitemaps alone to be found, so discovery is slow and demand stays low.

Every Google fetch counts against the same host. Ads and Shopping fetches share host capacity with Googlebot. A merchant feed or ad landing pages hitting a slow server can reduce how much is crawled for Search.

○ Who reads robots.txt, and who doesn't

WHO FETCHES	ROBOTS.TXT	WHAT WAS SAID ON DAY 1 STAGE
Googlebot and Google's other automated crawlers	Obedied	robots.txt is how site owners opt out of Google's crawling.
Contractual crawlers	May ignore	They crawl sites whose owners agreed to it. DOCS AdsBot ignores the * group, with the publisher's permission.
User-triggered fetchers	Not checked	A user asked for the fetch, as with a translation service.
AI agents acting for a user	Not checked	Same category: an agent that obeyed a block couldn't finish a purchase.
Other search and AI crawlers	Try to obey	Those that ignore it and cause a nuisance are a scraping problem.

STAGE A Google panelist: much of robots.txt is written for search engines. A search engine should never add items to a cart; an agent acting for a user probably should.

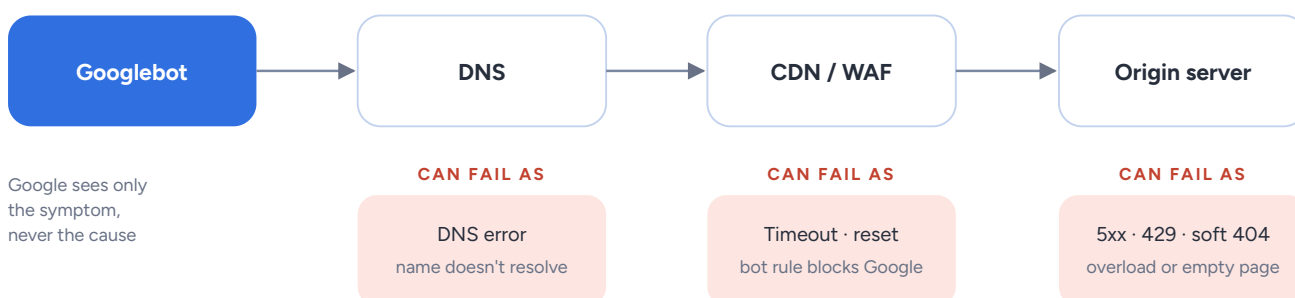
06

SECTION 06 • HOW CRAWLING ERRORS AFFECT SEARCH

Crawl errors: what Google sees when it can't reach you

The most damaging crawl errors are the ones site owners never see. A firewall that blocks Googlebot shows up on Google's side only as failed connections — and Google reacts by crawling less.

STAGE Gary Illyes: DNS problems and firewalls are blind spots. Google doesn't know a firewall is the cause, it only sees requests fail. He called soft 404s one of the biggest problems on the internet right now for crawling and for showing up in Search.



Where a fetch can fail. Each layer between Googlebot and your page can fail in a way that looks like a sick server.

Status codes, class by class

CLASS	WHAT CHERRY PROMMAWIN EXPLAINED STAGE
1xx Informational	Only says the request arrived and more is coming; no meaning of its own for crawling (<i>not in docs</i>).
2xx Success	200 : what comes back is eligible for indexing — though, as Gary Illyes put it, 200 only means the server <i>believes</i> it did what was asked. 204 No Content: nothing to index.
3xx Redirection	Followed to the new location, permanent (301 , 308) or temporary (302 , 307); indexing depends on what the target returns. 304 : unchanged since the last visit.
4xx Client error	404 and 410 : nothing here, not indexable. 403 : no permission, so nothing to index. 429 is the exception: it means "slow down", and Google does.
5xx Server error	Crawling slows, because the server, and often the whole site, can't serve requests, and Google doesn't want to break it.

DOCS Google's crawlers follow up to 10 redirect hops by default; content served by the redirecting URL is ignored and only the final target is processed.

○ How Google reacts to each response DOCS

WHAT GOOGLE GETS	TREATED AS	EFFECT
5xx server error	Overload signal	Crawling slows; indexed URLs are kept for a while, "but eventually dropped".
429 Too Many Requests	Overload signal	Crawling slows, like 5xx .
4xx other than 429	Normal "not found"	"No effect on crawl rate"; the URL leaves the index.
200 with empty or error page	Soft 404	Excluded from the index — and still crawled, wasting budget.
Timeout, reset, DNS failure	Network / DNS error	Treated like 5xx: crawling slows at once, and indexed URLs that stay unreachable drop out of the index within days.
robots.txt returns 5xx	Rules unreadable	Crawling stops for 12 hours, then the cached copy is used for up to 30 days.

○ Login walls, paywalls and pay-per-crawl

STAGE Gary Illyes: Google has recently seen more 403 responses to its crawlers, which he described as "authentication required". It treats them as client errors, technically like 404 or 410, and drops the pages from Search and its AI features. 402 Payment Required is no different, because Googlebot can't pay: "402 will just mean 404 to us". The uptick in 402 s is even more recent than the one in 403 s (*not in docs*).

ANALYSIS

Never answer verified Googlebot with 401, 402 or 403 from a login wall, a paywall or a pay-per-crawl setup: the pages leave the index. Google's status code page also says not to use 401 or 403 to limit crawling. To slow Google down for a while, return 429 or 503.

○ DNS, networks and CDNs

- **STAGE** Gary Illyes called DNS and network errors very common, and DNS problems very pesky to debug. Both can get a site removed from Search and, with it, from every feature that depends on Search; DNS errors do so very aggressively. Search Console alerts you.
- **STAGE** Most network errors happen between your origin and Google's data centres, invisible to both sides (*not in docs*). Timeouts usually start close to the site: a firewall, the CDN, the host or the DNS provider.
- **STAGE** A rise in HTTP errors can be a CDN throttling crawlers by injecting 429 or 503 responses on the way between Google and your site. CDN captcha challenges often return 200; Googlebot doesn't solve them, so indexing classes the page as a soft 404. **ANALYSIS** Day 2 added that such pages can also be clustered as duplicates. Either way, the real pages drop out.
- **STAGE** What to do first: ask the host or CDN whether they changed anything, check for new firewall rules, then ask in the CDN's forum. Google can't see these errors, so it can't help with them.

○ HTTP/3

STAGE From my notes, session not recorded: Googlebot does not support HTTP/3 today. DOCS Consistent: Google's crawlers use HTTP/1.1 and HTTP/2, pick whichever crawls best, and may switch between sessions. HTTP/2 can save server resources but gives no ranking benefit.

ANALYSIS

This doesn't mean "don't use HTTP/3" — browsers benefit from it, and servers offer it alongside HTTP/2 and HTTP/1.1. The rule is: never run a host that only works over HTTP/3, and make sure CDN or firewall rules written for HTTP/3 traffic don't break the protocols Googlebot uses.

○ How to find these errors

1. Search Console → **Settings** → **Crawl stats** → **Host status**: robots.txt fetch, DNS resolution and server connectivity. Any red in the last 90 days deserves a ticket.
2. In the same report, break requests down **by response**. A rising share of **5xx** or **429** means Google is being told to slow down.
3. Ask whoever runs the CDN or WAF for blocks on **Google's published crawler IP ranges**. Verify Googlebot by reverse DNS, never by user agent alone.
4. For temporary blocks, return **503** or **429** — never a **200** page with a captcha or error message, which becomes a soft 404 and can replace your real content in the index.
5. Review **Soft 404** in the page indexing report, where Gary Illyes said to look them up; Crawl Stats is for the other crawl issues. On local and service sites, thin location pages with little unique content are the usual cause.



07

SECTION 07 • HOW GOOGLE INTERPRETS ROBOTS.TXT • LIGHTNING SESSION B

robots.txt, read the way Google reads it

robots.txt has only three rules, yet most mistakes come from three things people get wrong: a crawler obeys only one group, the longest matching path wins, and `$` anchors the whole URL — parameters included.

What robots.txt is, and what it isn't

STAGE Google's talk started in 1994, when bots were crashing servers and Martijn Koster proposed a text file at the root of a site with rules for automated clients. Google supported it from its first crawl in 1996, as a Stanford PhD project. Today it is an IETF standard, RFC 9309, officially the Robots Exclusion Protocol, and basically all major crawlers follow it; Google does because anyone who wants to opt out of crawling should be able to. It controls which clients may access what. It has nothing to do with how the content is used.

Not a security measure

STAGE The file sits at a predictable public address, so anyone can read the paths it disallows. Protect a secret folder with authentication, or keep it off the internet.

Three rules, one place

STAGE Always at the root of the host. Only `user-agent` (a named crawler, or `*` for all), `disallow` and `allow`. Everything is allowed by default, so `allow` only re-opens a path inside a disallowed one.

Comments

STAGE `#` starts a comment, and a line a parser can't read is skipped anyway, as the standard requires. To show why comments help, the talk pointed to the file of `thebestfriedchickenever.com`: mostly an ASCII-art drawing in comment lines (*not in docs*).

STAGE Google called the format extremely forgiving: a typo in a path only blocks the wrong path, a typo in a rule name makes Google ignore that line, and the rest of the file still counts (*not in docs*). **ANALYSIS** Google's open-source parser even accepts common misspellings of `disallow` and `user-agent`, though not of `allow`. Other crawlers may be stricter: spell rule names correctly.

The puzzle from the slide

```
# Let's craft a robots.txt!
```

```
user-agent: *
disallow: /
allow: /$
```

```
user-agent: bingbot
user-agent: googlebot
disallow: /
allow: /$
allow: /politics/
allow: /*/live/
```

```
user-agent: google-extended
disallow: /subscriptions/
allow: /
```

DOCS The rules that decide it:

- One group per crawler.** Several `user-agent` lines can share a group. A crawler follows only the most specific group naming it — so Googlebot ignores the `*` group here entirely.
- Longest path wins.** Within the group, the rule with the longest matching path applies.
- Ties go to allow.** If an allow and a disallow match equally, the less restrictive wins.
- * and \$.** `*` matches any characters; `$` marks the end of the URL.

What Googlebot may crawl under this file **ANALYSIS**

URL	DECIDING RULE	GOOGLEBOT
/	allow: /\$	Allowed
/?utm_source=newsletter	disallow: / — /\$ needs the URL to end right after /	Blocked
/index.html	disallow: /	Blocked
/politics/eu-vote	allow: /politics/ (longer than /)	Allowed
/politics	disallow: / — no trailing slash, no match	Blocked
/sports/live/	allow: /*/live/	Allowed
/live/	disallow: / — pattern needs a segment before /live/	Blocked
/sports/live-score	disallow: / — no /live/ with a slash	Blocked

Every other crawler falls into the `*` group and may fetch only the bare homepage. `google-extended` is not a crawler — see below.

STAGE The talk's own walk-through used the same tools the other way round: `disallow: /*/live/` blocks `/science/live/` and `/sports/live/`, and `allow: /science/live/` re-opens that one path. `user-agent: *`, `disallow: /` and `allow: /$` let unnamed crawlers fetch only the homepage: `/$` means only the root, `/cats$` exactly `/cats`.

Groups: one per crawler, never added together

STAGE A `user-agent` line with its rules forms a group, and one group can name several crawlers, such as Googlebot and Bingbot. Give a crawler its own group only when it needs rules the `*` group shouldn't grant everyone. To give one crawler an extra rule on top of another group's rules, add one more group for it: Google's spec merges all groups that name the same user agent.

```
# The quiz answer
user-agent: googlebot
disallow: /staging/
allow: /staging/preview/
```

STAGE The talk ended with a quiz. Its answer here: one Googlebot group holding both rules, not an allow in the `*` group and not a Googlebot group with only the allow.

DOCS Why: a crawler follows only the most specific group naming it. An allow in the `*` group opens the path to every crawler; a Googlebot group with only the allow frees Googlebot from the `*` group's `disallow: /staging/`, so all of `/staging/` opens to it.

The unknown-line trap

STAGE The real trouble, Google's talk said, is an unknown line between two `user-agent` lines. Should the first crawler inherit the rules that follow? While writing RFC 9309, Google asked people, and opinions split roughly 50-50 (*not in docs*). How the question was settled is unclear in the recording.

SLIDE A later slide of Google's talk showed how two crawlers read such a file: "Different crawlers behave differently! ... Unpredictability is never a good time."

HOW GOOGLEBOT READS IT

```
User-agent: bingbot
Content-signal: all-access=yes
User-agent: googlebot
Disallow: /
```

Both bingbot and googlebot blocked

Unknown line ignored; one group, one rule

HOW BINGBOT READS IT

```
User-agent: bingbot
Content-signal: all-access=yes
User-agent: googlebot
Disallow: /
```

bingbot not blocked

Group closed early; bingbot ends up with no rules

Same file, two readings. Bars show how each crawler groups the lines, per the talk's slide.

ANALYSIS

`Content-Signal` lines are now common because a large CDN provider adds them to its managed robots.txt files. Google supports only `user-agent`, `allow`, `disallow` and `sitemap` and ignores the rest. Never place any other directive between `user-agent` lines: put it after a group's rules, and test the file with each search engine's tools (Bing's robots.txt tester, Google's open-source robots.txt parser).

Google-Extended, crawl-delay and llms.txt

DOCS Google-Extended is a control token, not a crawler. It decides whether content may train future Gemini models and ground Gemini apps and Vertex AI, and it "does not impact a site's inclusion in Google Search nor is it used as a ranking signal".

STAGE The talk: disallow a directory or the whole site for Google-Extended and Google keeps it out of Gemini model training, and respects that. Other companies have similar tokens, such as Apple's Applebot-Extended (*that part not in docs*). The speaker believed Google was the first to offer such an opt-out. **ANALYSIS** Not quite: OpenAI documented how to block GPTBot in August 2023, before Google-Extended arrived on 28 September 2023. What was new was a token that separates AI-training use from Search for pages Google's existing crawlers already fetch.

crawl-delay

DOCS Not supported by Googlebot; silently ignored. robots.txt is cached for up to 24 hours, and only the first 500 KiB is read.

llms.txt

SLIDE "LLMS.txt isn't necessary." **STAGE** Speakers at the event were openly dismissive. **DOCS** Such files "neither harm nor help" — Google Search ignores them.

The robots.txt report

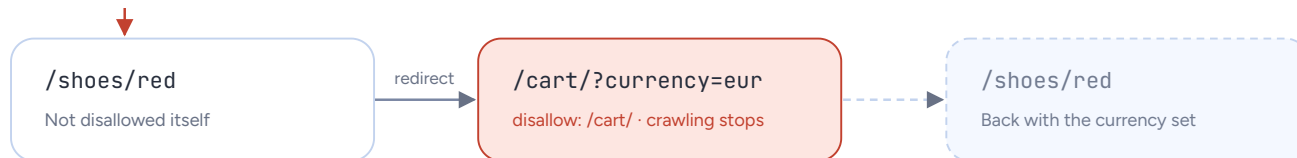
STAGE Search Console's robots.txt report shows the file as Google last fetched it, with a version history and the errors and successes of each fetch. It uses the parser Google open-sourced at github.com/google/robotstxt. Use it to catch CDNs that change your robots.txt without telling you, and hosts that cloak the file, which happens more often than people think (*the last two points not in docs*).

DOCS The report covers the top 20 hosts of a Domain property, with fetch status, warnings and errors. To test one URL, Google points to the URL Inspection tool and its open-source robots.txt library.

○ Lightning session B: when a redirect meets a blocked URL

STAGE Dave Smart opened Lightning session B with the pitfalls he sees most. First: groups are not additive. If the `*` group blocks `/goats/` and `/cows/` and you add a Googlebot group that blocks `/dogs/`, only that group applies to Googlebot, so it is free to crawl the goats and the cows.

SEARCH CONSOLE: BLOCKED BY ROBOTS.TXT



Every URL in the chain is checked against robots.txt; the report names only the first one.

Blocked by a URL you never see. Dave Smart's example, redrawn with illustrative paths: a JavaScript redirect through the cart to set the local currency.

- **STAGE** robots.txt is checked for every URL in a redirect chain, and crawling stops at the first blocked one (*not in docs*). Search Console reports the block only on the first URL, which is confusing: the URL shown as blocked isn't disallowed itself.
- **STAGE** It applies to every kind of redirect, not only JavaScript: a hop through an external authorisation service blocked by its own robots.txt, content that moved through several URLs over the years until one of them got blocked, or a redirect served only to Googlebot's user agent (*not in docs*).
- **STAGE** So when Search Console says a URL is blocked by robots.txt and the file doesn't block it, check whether the URL redirects, and test every URL in the chain.

08

SECTION 08 • CHERRY PROMMAWIN, SEARCH RELATIONS

Crawl budget

Crawl budget is two separate limits: how much your server can take, and how much Google wants. Speed raises the first; quality and popularity raise the second. Most sites only ever hit the second.

STAGE In one line: crawl budget is the attention span Google gives a website — and better performance increases it. More formally, it is the finite amount of crawling resources Google gives a site, which decides how many of its pages are found and how often they are revisited.

CRAWL RATE LIMIT • HOSTLOAD

How much your server can take

Host-wide, shared by all Google crawlers

- ↓ Connect time rising
- ↓ Time to first byte rising
- ↓ HTTP 429 or 5xx responses

+

CRAWL DEMAND

How much Google wants your URLs

Unknown URLs inherit from their parent path

- ↑ Quality of the site
- ↑ How often URLs change
- ↑ Popularity on the web

=

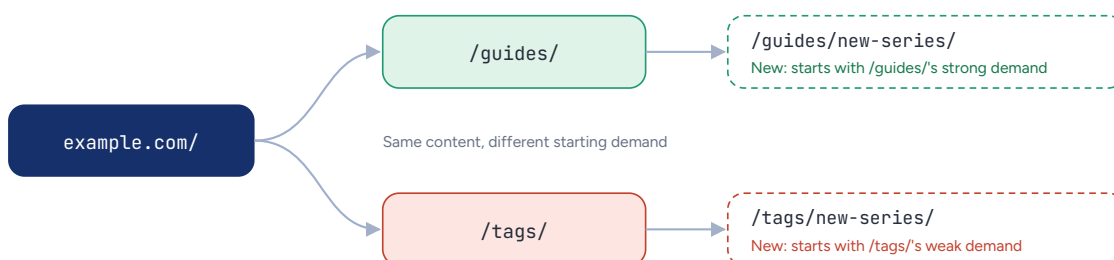
Crawl budget

The formula. From the session slides: "Crawl Budget = Crawl Rate Limit & Crawl Demand".

- STAGE** Hostload is a proxy for how many requests per second the connection can take. It applies **per host**, not per site: your CDN, app, subdomains and `www` host may each be a different host.
- STAGE** The quality behind crawl demand is the quality of the site as a whole, not of one page (*not in docs*).
ANALYSIS Read it as site quality setting the baseline; Google's guide still lists page-level quality among the demand factors.

Quality is inherited by path

SLIDE If quality or popularity is unknown, the parent root's aggregate quality or popularity is used, then that path's parent's, and so on (*not in docs*).



Illustrative example. How path inheritance could play out for new content, based on the slide's rule. Paths are hypothetical.

ANALYSIS

Put new high-value content where Google already trusts you. A new folder under a section Google rates poorly starts with that section's low demand.

Who needs to care

DOCS Google's crawl budget guide is written for sites with **1 million+** unique pages changing weekly, **10,000+** pages changing daily, or many URLs stuck in "Discovered – currently not indexed". **ANALYSIS** On a smaller site, slow indexing is almost always a demand problem (quality), not a capacity problem.

STAGE Asked whether crawl budget is still a priority in 2026: not every site needs to worry, and that has always been so. A site with fewer than a few thousand URLs likely has no problem, and a larger one doesn't necessarily have one either.

ANALYSIS Between that and the guide's million-page threshold, check Crawl Stats before blaming crawl budget for slow indexing.

What burns it: the six faceted-URL traps

SLIDE Server errors unrelated to load, useless pages and resources, and infinite URL spaces. Faceted navigation was the worked example. **STAGE** URLs that differ are different URLs to Google, crawled even when they lead to the same content: calendars and endless page versions burn budget.

TRAP	EXAMPLE FROM THE SLIDE	FIX
1 • Same facets, different order	<code>/shoes/color-red/size-10</code> <code>/shoes/size-10/color-red</code>	Always build facet URLs in one fixed order
2 • Impossible combinations	<code>/laptops/brand-apple/os-windows</code>	Return <code>404</code> when a combination has no results
3 • Too many facets	<code>/cameras/type-dslr/brand-canon/...</code>	Make one or two facet levels crawlable; block deeper ones
4 • Facets on pagination	<code>/phones/.../page-2/sort-price-asc</code>	Fixed parameter order; block sort parameters
5 • Facets + internal search	<code>/search?q=blender&brand=ninja&...</code>	Disallow internal search URLs in robots.txt
6 • Default-value facets	<code>/televisions/size-55inch/resolution-4k</code>	Drop default values; link only to the normal form

DOCS Google's faceted navigation guide prefers prevention: block filter URLs in robots.txt and keep crawlable only item pages plus one unfiltered listing page — or use URL fragments, which Google generally doesn't crawl. `rel="canonical"` and `nofollow` are weaker, slower options.

○ How to manage it

01

Use HTTP caching.

Serve **304** for unchanged pages; ETag preferred.

02

Have good navigation.

Clear internal links for discovery and priority.

03

Restrict faceted and action URLs — add to cart, wishlist, print, sort.

04

Improve or remove useless content: low-quality, spam, duplicate and soft-error pages. Return **404 / 410** for removed pages.

○ What each directive costs

DIRECTIVE	CRAWL BUDGET SLIDE	THE CATCH DOCS ANALYSIS
disallow	Not consumed — never fetched	Can still be indexed without content if linked, and Google can't see its noindex . Not for temporarily shifting budget.
noindex	Consumed — page must be fetched	The right tool to keep a page out of the index; the wrong tool to save crawl.
nofollow	Can be consumed	A hint, not a block. Google doesn't crawl through the nofollow link itself, but crawls the target when it finds it through other links or sitemaps.
crawl-delay	Consumed — ignored	No throttling effect. To slow Google temporarily, return 503 or 429 .

STAGE Status codes don't all cost the same. Every **2xx** fetch consumes crawl budget; **4xx** responses don't affect it, because pages come and go on the web. **ANALYSIS** A 404 is still a fetch, so read the point this way: a 4xx doesn't slow crawling, and Google crawls a 404 URL less often over time. **429** is the exception and slows crawling like a **5xx**.

QUESTIONS SENT IN BEFORE THE EVENT

Does AI Mode mean more crawling? **STAGE** No: AI Mode sits on top of Google's existing crawling infrastructure, so Google doesn't crawl a site more because of it. Expect more crawling overall, though, from the many other services, AI included, that now crawl the web.

Does crawl frequency affect rankings? **STAGE** "If your crawl rate is increased, that doesn't mean that you would rank better." Frequent crawling is a consequence, not a cause: it tends to signal interest in the site and good content.

How do I check? **STAGE** In Crawl Stats, by response and by file type: most crawling going to images on a site with no images worth crawling is an anomaly. It also separates discovery (new URLs) from refresh (known URLs) and drills down to the problem URLs. Look now and then; don't obsess.

09

SECTION 09 • LIGHTNING SESSIONS A AND C

Lightning talks from the community

Google picked these seven-minute talks from close to 200 submissions, because the SEO community holds knowledge worth sharing. They are community views, not Google's: the labels show what Google's documentation backs.

Session A: automation and AI

AI THAT APPLIES YOUR RULES

James Powley

STAGE Break a deliverable into steps and use AI only where it adds scale; people set the strategy and the scoring. A competitor keyword analysis (26 competitors, 79 data points per keyword) fell from three days to a few hours, and a 3,000-post blog audit came back grouped, flagged for cannibalisation and prioritised. On the AI: "it didn't decide what good looked like. We did that."

LLM ERRORS ON SEARCH DATA

Rafael Kovashikawa

STAGE A chatbot called an average position moving from 8 to 9 an improvement. His rules: do "not let the model grade its own homework"; every figure needs a receipt, traceable to the data request ("If there is no receipt, there is no reimbursement."); give the model your metric definitions as fixed context. Polished prose is no proof of correct analysis: borrowing Pascal's line on makers of false windows, he said LLMs sometimes don't aim for the most accurate figures. **DOCS** In Search Console, position 1 is the top.

AN LLM FIX LOOP FOR STRUCTURED DATA

A community demo

STAGE A script runs Google's Rich Results Test, by URL for public pages and by pasted code for pages Google can't fetch, gives a large model the markup, the errors and Google's documentation, and re-tests the new JSON-LD, with at most three attempts. A small, cheap model picked at most two keywords per page, never brand or navigation terms. Errors on the test page included an empty name and a price of zero; warnings were treated as not critical (a largely unintelligible recording). **DOCS** Only critical issues stop a rich result. Code mode still needs the page's resources reachable without a login, and merchant listings need a name and a price above zero. **ANALYSIS** A passing test proves the JSON-LD is valid, not that its values are true.

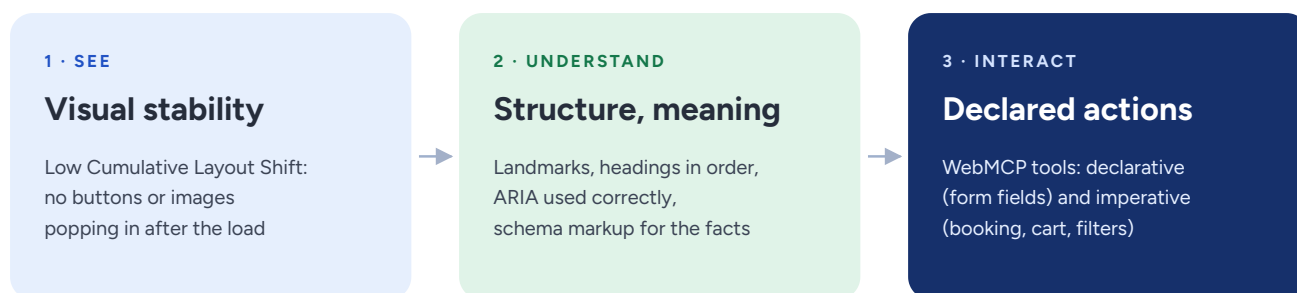
AUTOMATING MONTHLY REPORTS

Kira Breuer

STAGE Monthly reports took each SEO manager a full workday, and nobody read them. A non-developer automated them with Python and AI coding help: data in, a script, an AI summary, a dashboard. Start tiny, read the code the AI writes: "you don't need to be an expert; you just need to try." **DOCS** The Search Console API is free, with usage limits, and needs OAuth2 credentials.

Preparing sites for humans and AI agents • Carlos Ortega

STAGE The web is now visited by people and AI agents, and agents understand a page through a screenshot, the DOM and the accessibility tree. **DOCS** Google's guide for generative AI features describes browser agents the same way and recommends semantic HTML.



Agent readiness in three layers. Carlos Ortega's model. Google's docs back the accessibility-tree part; WebMCP is a proposed standard.

- **STAGE** ARIA is a set of attributes, not a language: a `div` used as an "add to favourites" button gets `role="button"`, an `aria-label` and `aria-pressed`. No ARIA is better than bad ARIA.
- **STAGE** WebMCP lets a site declare actions for agents instead of being scraped; not yet in all browsers, but testable through browser flags. **DOCS** Chrome describes it as a proposed web standard, testable in an origin trial from Chrome 149 or behind a flag.
- **STAGE** Don't publish markdown copies of pages for agents: they are duplicates (he also called them a possible source of cloaking, an uncertain word in the recordings), and models read HTML, CSS and JavaScript anyway. `llms.txt` is not necessary: in a summer 2026 study, heard as Ahrefs', 97% of almost 40,000 sites had no AI agent hits on the file. **ANALYSIS** Ahrefs' published study fits: about 38,000 of 137,210 domains had the file, and 97% of those files got no requests in May 2026; the rest got about 20 each, not "two hits a month".
- **STAGE** Schema markup tells an agent which number is the price (*not in docs*). **ANALYSIS** Google says structured data is not required for generative AI search; use it for rich results and clear data, not as an AI lever.
- **DOCS** Headings out of order don't matter to Google Search. **ANALYSIS** Keep a logical hierarchy anyway, for screen readers and agents that read the accessibility tree.

GEO: its numbers and its myths • Thiago Pojda

- **STAGE** SEO turned everything into a metric because "an idea doesn't get you budget approval; a number does". Not all of it was wrong: a site still has to be crawlable and visible, and much of that work is plumbing, not gaming.
- **STAGE** A study found sites with JSON-LD shown much more often, and the industry added schema everywhere; a later controlled study that only added markup found a negative correlation. The markup marked teams doing everything else right (*not in docs*).
- **STAGE** Listicles earn more AI mentions because LLMs mostly check that a thing exists, not that the content is honest; the industry is overusing them as it once overused links.
- **STAGE** A third-party study, heard as Peec AI's: 43% of prompts produced fan-out queries in another language than the user's (*not in docs*). A Spanish prompt about places to eat in Barcelona fanned out entirely in English. **ANALYSIS** Peec AI's published analysis (February 2026, over 10 million ChatGPT prompts) puts it differently: 43% of the fan-out searches for non-English prompts ran in English, and nearly 78% of non-English prompt runs had at least one English fan-out. The 43% is a share of fan-outs, not of prompts.
- **STAGE** The speaker argued for adopting the GEO label as a fresh start for the industry — the opposite of Google's view that morning.

AI search as a funnel STAGE

STAGE	THE QUESTION	MEASURE PROPOSED BY A COMMUNITY SPEAKER
Know	Can AI training bots reach the site?	"AI bot allow rate": share of AI training bots allowed (3 of 4 = 75%); aim for 100%
Association	Does the model link the brand to its topics before it searches?	"Brand association rate": ask, with browsing off, for ten topics; repeat over time
Search	Does the brand appear in the AI's fan-out queries?	"Brand co-occurrence rate": share of fan-out queries naming the brand
Retrieval	Does the AI fetch your pages to ground its answer?	If not, look for a crawling or indexing issue
Selection	Is the brand chosen in the answer?	Mentions; most tools measure only this last stage

STAGE Closing the session, a Google host called AI a tool that can be used well or badly. It can even generate content, which none of the talks featured, and it is fantastic for coding and analysis. The host advised always checking what AI tools do and focusing on what the business and its users need.

Session C: crawling

SIMILAR URLS

Tobias Schwarz

STAGE URLs that differ in a few characters point to duplicates, broken links and needless redirects; he found five kinds on well-known brand sites (table below). To find them, normalise the URL list, build a "uniform" URL (one scheme, lowercase, no `www` or `index.html`, sorted parameters) and sort by it. To prevent them, compute the expected URL for every request and redirect when it differs; test IDs with leading zeros too. ANALYSIS Prefer the redirect to an error page: it passes signals to the canonical.

TECHNICAL SEO IN AI SEARCH

Jovana Avramovic

STAGE More than half of internet traffic in 2025 was non-human, with no source named (*not in docs*). Search now happens everywhere and in conversation, and AI turns one query into many, in other languages too; technical SEO stays the foundation. Models, like people, remember the start and end of a list best: "It's relevant to put your most important information at the beginning or at the end." ANALYSIS A community heuristic, not something Google said; Google advises writing for people without chopping content. Her summary: "The target is not shifting; it is expanding."

Five kinds of similar URLs STAGE

WHAT DIFFERS	TWO URLs, ONE PAGE (ILLUSTRATIVE EXAMPLES)
Protocol or host	<code>http://example.com/shoes</code> · <code>https://www.example.com/shoes</code>
Capitalisation	<code>/Shoes/Red</code> · <code>/shoes/red</code>
Number of delimiters	<code>/shoes/red</code> · <code>/shoes//red/</code>
Encoding of a space	<code>/search?q=red%20shoes</code> · <code>/search?q=red+shoes</code>
Parameter order	<code>?size=10&color=red</code> · <code>?color=red&size=10</code>

STAGE They usually come from programming errors or hand-set links, but anyone can link to them from other sites, so harden the application against them (*not in docs*).

10

SECTION 10 • Q&A • JOHN MUELLER AND GARY ILLYES

The Q&A: crawling questions, answered

Questions from the room and from registration, paraphrased, now checked against two audio recordings of most of the panel. An answer names a speaker only where the recording shows who gave it; otherwise it says "a panelist" or "Google".

QUESTION

How does crawling for AI differ from crawling for Search?

STAGE A panelist: "my feeling is a lot of the AI crawlers are still a bit stupid." In logs, search crawlers follow what changes and what is valuable; many AI crawlers work through a site in order and take everything. Training mainly needs a very large number of tokens, wherever they come from (*not in docs*). Can optimising for AI hurt classic search? Yes, a panelist said, pointing to questionable GEO advice online.

QUESTION

Does lms.txt matter? PageSpeed Insights and Lighthouse check for it.

STAGE Gary Illyes: not for Google Search, and he doesn't expect it to, though he has been wrong before; it matters to some people, hence the Lighthouse check. A panelist: "Just because maybe something will change in the future doesn't mean you should take action now." Most such files come from an SEO plugin checkbox, and "it's designed for allegedly intelligent systems" that should be able to parse a website. And cats.txt, a satire, gets more requests than lms.txt in some logs: crawlers fetch whatever they find. **DOCS** Lighthouse marks its lms.txt audit not applicable when the file is missing.

QUESTION

How are huge real-time sites, such as sports sites, crawled?

STAGE With no special handling. When URLs change often or are useful to users, Google raises crawl demand and tries to raise its crawl capacity for the site, the same logic as for small sites. If the server can't cope, Google backs off again. And demand isn't only Search's: when Search, Google Ads or another product wants enough of a site's URLs, Google raises demand and crawls up to it, as far as capacity allows.

QUESTION

How do I spot crawl waste in my logs?

STAGE Logs alone can't show it, because they don't show where your crawl limit lies (*not in docs*). In Crawl Stats, check whether total crawling has plateaued and whether response times hold Googlebot back. Logs do pay off on sites with filters or complicated URLs: redirect parameter variants to the normalised URL, which costs crawl budget at first but leaves a clean slate. Many complaints turn out to be a plugin, such as a WordPress calendar adding calendar parameters to every page's URLs (*not in docs*); Google crawls them until large samples show they are useless. Such URLs can also be handled at the web server, with a rule or a module in Apache, for example.

QUESTION

Is crawl budget slowing how fast new articles are found?

STAGE Gary Illyes: news sites rarely need to worry; Google crawls them aggressively. Measure the time from publishing to Googlebot's first crawl: flat or falling is fine, a rising trend needs work, and for fresh stories something like two hours is probably not great (*not in docs*). Then look in the logs for URLs crawled again and again, and disallow sections you never want crawled, such as `/ads`. John Mueller: Crawl Stats' discovery-versus-refresh split gives a rough view; if about half the crawling goes to discovery, finding new pages is not the problem. **ANALYSIS** Freed budget only moves on sites that hit their capacity limit.

QUESTION

On a site of 100 million pages, is a crawl drop about demand or capacity?

STAGE There is no easy way to tell, but most of the time a capacity-limit drop is an abrupt step down: hostload going from 10 to 5, say. Look at how many connections Googlebot opens to the site: if that number fell, the capacity limit changed. **ANALYSIS** The panelist put the step as five requests per second; Google's guide counts the limit in parallel connections and how long they stay open, not in requests per second.

QUESTION

What will Search Console report on AI features?

STAGE A panelist expected more AI reporting, without a timeline or a promise. Google said the reports shouldn't simply mirror classic results, because people learn more before they click, so it first has to find which data is useful. Query data is hard: AI questions don't map back to keywords, and the data would have to be grouped to protect privacy.

QUESTION

Will crawl stats come to the Search Console API?

STAGE No plans known: it would be a new part of the API, while a variation of the Performance report would be easier. Search Relations now talks with the Search Console team about the API more than before: with so many vibe-coded tools, the easy answer is to use the API, and a gap in it is annoying. To get a feature built, ask for it publicly and in volume.

QUESTION

How do I migrate without losing indexing?

STAGE Probably not with sitemaps: decide what matters to the business first. Listing the new URLs can't hurt, but it isn't the main tool. Google merged 12 or 14 language blogs, a Help Center and its developer site into one. First it protected the popular URLs (*not in docs*); where two language versions duplicated each other, both old URLs got one target path. A fun fact: JavaScript was used, the only option the person doing it had (whether for the redirects themselves or for the mapping is not fully clear) (*not in docs*). **ANALYSIS** That was the move to Google Search Central in November 2020, about six years before the event. Don't copy the JavaScript: Google sees such redirects only after rendering, and asks for server-side 301 or 308 redirects where possible. **STAGE** And a sitemap has no sweet spot: list every URL you want indexed.

QUESTION

Will Google offer separate AI user agents?

STAGE Nearly all mainstream AI systems already use their own. A panelist doubted per-bot policies make sense yet, and called blocking all AI crawlers a personal choice. For Google's AI training, use Google-Extended. An unknown crawler in your logs: search its user agent for its token (*not in docs*). Default-deny is "a bad pattern, because you don't know what you're blocking." **ANALYSIS** Google-Extended is a usage token, not a crawler: it doesn't stop Googlebot.

QUESTION

What if paginated pages become one load-more page?

STAGE Thousands of paginated category pages replaced by a JavaScript load-more button: should the old pages redirect to page 1? A panelist called pagination one of the trickiest things in web development, and the switch "kind of dangerous, depending on what you want to achieve": "Googlebot is not running around clicking on buttons". Without crawlable links to the further pages, Google won't see them at all. **DOCS** Each page needs its own URL and a plain `<a href>` to the next; don't canonicalise to page 1; `rel=next/prev` is no longer used.

QUESTION

How do I make crawlers behave, beyond robots.txt?

STAGE Research, malware-scanning and privacy crawlers are a separate, generally useful category. Google is working on crawler best practices and offering them an exemption, apparently because they sometimes need to ignore robots.txt or probe URLs other crawlers wouldn't touch (a best reading of the recording) (*not in docs*). **ANALYSIS** This matches a public IETF draft, "Crawler best practices": a draft, not Google documentation.



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SECTION 11 • FROM MY NOTES

From the floor

Points said on stage that neither my notes nor the recordings tie to a session, checked against Google's documentation where possible.

No duplicate content penalty

STAGE Gary Illyes: there is no such thing as a duplicate content penalty. **PRESS** John Mueller, quoted by Search Engine Journal in April 2026: "no penalty or ranking demotion if you have multiple URLs going to the same content". **ANALYSIS** The real costs: Google may choose a canonical you didn't want, and every copy is crawled. Scraped content is a separate spam issue.

Images matter

STAGE Images are important. **SLIDE** One in six AI Mode searches is multimodal, and a slide listed resolution, colour and associated text as image ranking signals. **ANALYSIS** Original photos, descriptive file names, alt text and captions now feed AI answers too.

Accessibility

STAGE Check ARIA and accessibility. **ANALYSIS** Google hasn't called ARIA a ranking signal. The case is that AI agents and assistive tools read pages through the same structure: real buttons, labelled forms, semantic headings.

Rich results

STAGE Check and focus on rich results for Google (*not in docs*). **DOCS** Search Console separates critical issues, which stop an item from showing as a rich result, from non-critical ones under "Improve item appearance".

SEO is more than content

STAGE SEO has many parts beyond content. **SLIDE** "Optimizing for people is optimizing for Generative AI Search."



Is SEO dead? Google's answer on stage was no: AI features run on the same crawl, index and ranking systems.

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SECTION 12 • TAKE IT BACK TO THE OFFICE

Day 1 audit checklist

Nineteen checks that turn Day 1 into work. Most take minutes in Search Console and one server log file.

ACCESS AND ERRORS

- ☐ Crawl stats → Host status shows no DNS, robots.txt or connectivity failures in the last 90 days
- ☐ Crawl stats by response: the share of 5xx and 429 is flat or falling
- ☐ CDN / WAF rules don't block Google's published crawler IP ranges; Googlebot verified by reverse DNS
- ☐ Maintenance, rate limits and bot challenges return 503 or 429, never a 200 error or captcha page
- ☐ Verified Googlebot never gets 401, 402 or 403 from a login wall, paywall or pay-per-crawl setup
- ☐ Every "Soft 404" in the Pages report is improved, merged, or returned as 404 / 410
- ☐ Every host answers over HTTP/1.1 and HTTP/2, whatever the HTTP/3 setup for browsers

ROBOTS.TXT

- ☐ Each crawler gets the rules you expect: only its most specific group counts, and groups don't add up
- ☐ No secret or private path is listed in robots.txt; private areas sit behind authentication
- ☐ No unknown lines (Content-Signal, crawl-delay) sit between user-agent lines
- ☐ The homepage with tracking parameters isn't accidentally blocked by a /\$ rule
- ☐ Google-Extended is set deliberately, knowing it governs Gemini, not Search
- ☐ The robots.txt report's version history shows no change you didn't make, for example by a CDN
- ☐ Every URL reported as blocked by robots.txt has its redirect chain tested, hop by hop

CRAWL EFFICIENCY AND AI EXPOSURE

- ☐ Filter, sort, internal search and action URLs are blocked or normalised; logs show Googlebot spending most requests on pages you want indexed
- ☐ URL variants (case, slashes, parameter order) redirect to one expected URL
- ☐ Load-more and infinite-scroll lists keep a crawlable <a href> to every page of the series
- ☐ The time from publishing a URL to Googlebot's first crawl is tracked, and flat or falling
- ☐ The Generative AI performance report is reviewed before any change to the Search generative AI setting

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SECTION 13 • STAY IN TOUCH

Help complete this guide

This guide covers what I could capture in the room. If you heard something on Day 1 that isn't written here, tell me and I'll add it, so everyone who attended ends up with the full picture.

01 • ADD WHAT I MISSED

Heard something that isn't here?

A second attendee's recording and six audio recordings filled most of the first edition's gaps. These are still missing:

- Lightning session B after Dave Smart's talk
- What would you do?: Crawling Edition
- The opening minutes of the crawl budget talk

Send me the point, who said it, and a photo of the slide if you have one. Corrections are just as welcome.

Each addition is labelled and checked against Google's documentation before it goes in, and credited to you by name if you like.

02 • THE FULL KNOWLEDGE BASE

Every Day 1 point, in the web edition

This guide is a selection. The web edition holds all of Day 1's points, more than 540, with topic pages, session pages, a search and the verification status of every point. It is a folder that opens offline from [index.html](#).

An agent-ready version of the same data lets AI tools answer questions about the event, with every answer traceable to its slide, session or document.

Message me and I'll share both.

LET'S KEEP IN CONTACT

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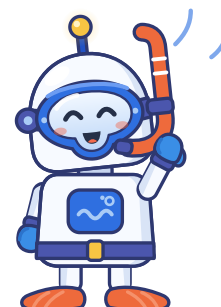
Message me with additions, corrections, or to get the web edition.



WHATSAPP



LINKEDIN



REFERENCES

Sources

This guide is built from 45 photos of the slides, the printed agenda, my notes from the room, a second attendee's recording of most of the day and six audio recordings. Every point labelled Docs was checked against these pages on 4 October 2026.

1. Search generative AI control — Google Search Console Help
2. Optimizing your website for generative AI features on Google Search — Google Search Central
3. Introducing Search Generative AI performance reports in Search Console — Search Central blog (3 June 2026)
4. A new profile to help publishers and creators highlight their work on Search — Google blog (4 June 2026)
5. 3 new ways we're improving Search profiles for publishers — Google blog (16 September 2026)
6. Manage your Search profile — Google Search Help
7. In-depth guide to how Google Search works — Google Search Central
8. A guide to Google Search ranking systems — Google Search Central
9. MUM: A new AI milestone for understanding information — Google blog (18 May 2021)
10. Creating helpful, reliable, people-first content — Google Search Central
11. Crawling infrastructure — Google
12. Overview of Google crawlers and fetchers (user agents) — Google
13. List of Google's common crawlers — Google
14. How Google interprets the robots.txt specification — Google
15. Optimize your crawl budget — Google
16. Managing crawling of faceted navigation URLs — Google
17. How HTTP status codes affect Google's crawlers — Google
18. Debug network and DNS errors for Google's crawlers — Google
19. Troubleshoot Google Search crawling errors — Google Search Central
20. Qualify your outbound links to Google — Google Search Central
21. Evolving "nofollow" – new ways to identify the nature of links — Search Central blog (10 September 2019)
22. Pagination, incremental page loading, and their impact on Google Search — Google Search Central
23. What is canonicalization — Google Search Central
24. Demystifying the "duplicate content penalty" — Search Central blog (12 September 2008)
25. Crawling December: HTTP caching — Search Central blog (9 December 2024)
26. Crawling December: CDNs and crawling — Search Central blog (24 December 2024)
27. Google Says It Can Handle Multiple URLs To The Same Content — Search Engine Journal (8 April 2026)
28. Search Central Deep Dive Europe 2026: Apparently we're going to Barcelona — Search Central blog (6 July 2026)
29. Introduction to robots.txt — Google Search Central
30. Build and submit a sitemap — Google Search Central
31. How is a Google Sitemap different from an HTML sitemap? — Search Central blog (26 September 2005)
32. Redirects and Google Search — Google Search Central
33. Understand the JavaScript SEO basics — Google Search Central
34. AI features and your website — Google Search Central
35. Search Engine Optimization (SEO) Starter Guide — Google Search Central
36. Robots Refresher: Future-proof Robots Exclusion Protocol — Search Central blog (28 March 2025)
37. Formalizing the Robots Exclusion Protocol Specification — Search Central blog (1 July 2019)
38. URL Inspection tool — Google Search Console Help
39. List of Google user-triggered fetchers — Google
40. Inside Googlebot: demystifying crawling, fetching, and the bytes we process — Search Central blog (31 March 2026)
41. How to specify a canonical URL with rel="canonical" and other methods — Google Search Central
42. How to move a site — Google Search Central
43. Merchant listing (Product, Offer) structured data — Google Search Central
44. Rich Results Test — Google Search Console Help
45. Google Search's guidance on using generative AI content on your website — Google Search Central

46. Page indexing report — Google Search Console Help
47. How AI powers great search results — Google blog (3 February 2022)
48. AI, personalization and the future of shopping — Google blog (3 March 2025)
49. Verify requests from Google crawlers and fetchers — Google
50. robots.txt report — Google Search Console Help
51. Minimize A/B testing impact in Google Search — Google Search Central
52. Automatically generating and ranking results — Google Search (How Search Works)
53. What are impressions, position, and clicks? — Google Search Console Help
54. Our approach to Search — Google Search (How Search Works)
55. Google Search's guidance about AI-generated content — Search Central blog (8 February 2023)
56. Generative AI performance report (Search) — Google Search Console Help
57. AI in Search is driving more queries and higher quality clicks — Google blog (6 August 2025)
58. Our Honest Results Policy — Google Search (How Search Works)
59. How ads work on Google Search — Google Search (How Search Works)
60. A new era for AI Search — Google blog (19 May 2026)
61. Google I/O 2026: Sundar Pichai's opening keynote — Google blog (19 May 2026)
62. How AI Mode is changing the way people search in the U.S. — Google blog (19 May 2026)
63. Q2 2026 earnings call: Remarks from our CEO — Google blog (22 July 2026)
64. Transformer: A Novel Neural Network Architecture for Language Understanding — Google Research blog (31 August 2017)
65. AI in Search: Going beyond information to intelligence — Google blog (20 May 2025)
66. 25 biggest moments in Search, from helpful images to AI — Google blog (6 September 2023)
67. Google rolls out AI Mode info monitoring to everyone globally — Relevant Audience (29 September 2026)
68. Rich result report overview — Google Search Console Help
69. Prerequisites — Google Search Console API documentation
70. Pricing — Google Search Console API documentation
71. Understanding Core Web Vitals and Google search results — Google Search Central
72. ARIA and HTML — web.dev (Google)
73. Content structure — web.dev (Google)
74. WebMCP — Chrome for Developers
75. List of Google's special-case crawlers — Google
76. Crawl Stats report — Google Search Console Help
77. Learn about sitemaps — Google Search Central
78. What Crawl Budget Means for Googlebot — Search Central blog (16 January 2017)
79. URL structure best practices for Google Search — Google Search Central
80. Myths and facts about crawling — Google
81. API Reference — Google Search Console API documentation
82. Goodbye Google Webmasters, hello Google Search Central — Search Central blog (11 November 2020)
83. llms.txt — Chrome for Developers
84. A new era of intelligence with Gemini 3 — Google blog (18 November 2025)
85. From the CEO: What's coming to YouTube in 2026 — YouTube Official Blog (21 January 2026)
86. ChatGPT searches in English, even when you don't — Peec AI
87. ChatGPT Search Often Switches To English In Fan-Out Queries: Report — Search Engine Journal (18 February 2026)
88. We Analyzed 137K Sites: 97% of llms.txt Files Never Get Read — Ahrefs (15 June 2026)
89. google/robotstxt — Google (open-source robots.txt parser and matcher library, GitHub)
90. An update on web publisher controls — The Keyword, Google blog (28 September 2023, Danielle Romain)
91. How to spot OpenAI's crawler bot and stop it slurping sites for training data — The Register (8 August 2023)

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